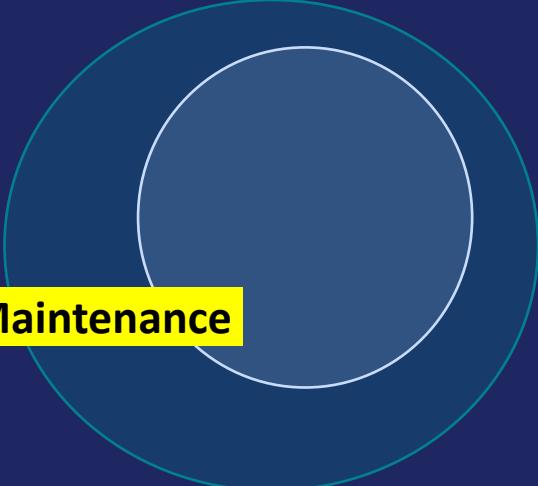




Exploring the Adoption Barriers of Digital Twin Technology for Maintenance Planning in Infrastructure Asset Management

Barriers, Interactions, and Strategic Pathways for UK Railway Infrastructure

MSc Design & Engineering | Network Rail Case Study | 2026

Methodology: Interpretivist qualitative case study | 9 semi-structured interviews | Reflexive thematic analysis

Presentation Structure

01

Research Context & Problem

Why Network Rail, why now

02

Literature Gap & RQs

What the literature misses

03

Methodology

Interpretivist case study design

04

Key Findings

Five themes, three original constructs

05

Adoption Framework (NRDTAF)

Five-principle strategic model

06

Contributions & Conclusions

Knowledge, recommendations, future work

01 Research Context

The problem

- ① UK railways manage ageing assets under growing demand (Whitmore et al., 2024)
- ② Digital Twin technology offers real-time monitoring, predictive maintenance, lifecycle optimisation
- ③ Adoption at Network Rail remains fragmented; mostly pilot-stage, not operational
- ④ The barrier literature is divided: Pregnolato et al. (2022) vs Whitmore et al. (2024)

Why Network Rail?

Critical case

HS1 DT programme, National Digital Twin Programme engagement

Scale & complexity

Legacy infrastructure, multi-stakeholder governance, safety regulation

Transferability

Findings applicable to comparable public-sector infrastructure organisations

02 Literature Gap & Research Questions

Gap 1

Empirical scarcity

No system-level case studies of DT adoption in UK heavy rail (Wu et al., 2025; GÜrdür Broo & Schooling, 2021)

Gap 2

Hierarchy contested

Technical-first frameworks (Kineber; Hakimi) vs. organisational-first (GÜrdür Broo; Opoku) — no resolution for rail

Gap 3

Maintenance underexplored

Existing studies focus on construction, energy, and water; maintenance planning as a DT domain is sparse (Zhao et al., 2022)

Research Questions

RQ1 What are the major barriers to Digital Twin adoption for maintenance planning at Network Rail, and how do they interact across technical, organisational, and governance dimensions?

RQ2 What organisational and policy interventions are most likely to address these barriers and support effective adoption at scale?

03 Methodology



Philosophy

Interpretivism
Socially constructed
knowledge; contextual
sensitivity over measurable
variables (Creswell & Creswell,
2018)



Design

Qualitative single-case study
Yin (2018): 'how' and 'why'
questions where phenomenon
cannot be separated from
context



Data Collection

9 semi-structured interviews
Operational → managerial →
strategic + 1 external DT
provider. Triangulated with
policy docs & literature



Analysis

Reflexive thematic analysis
Braun & Clarke (2022) six
phases. Inductive coding;
significance by consequence,
not frequency alone

Trustworthiness (Lincoln & Guba, 1985): Credibility · Transferability · Dependability · Confirmability

04 Key Findings: Five Interconnected Themes

T3

Governance, Regulatory & Financial

- Data ownership vacuum
- Regulatory: constraint + driver
- Investment paradox (original)

T4

Value Perception & Trust

- Overhyped strategically
- Underutilised operationally
- Qualified trust, not unconditional

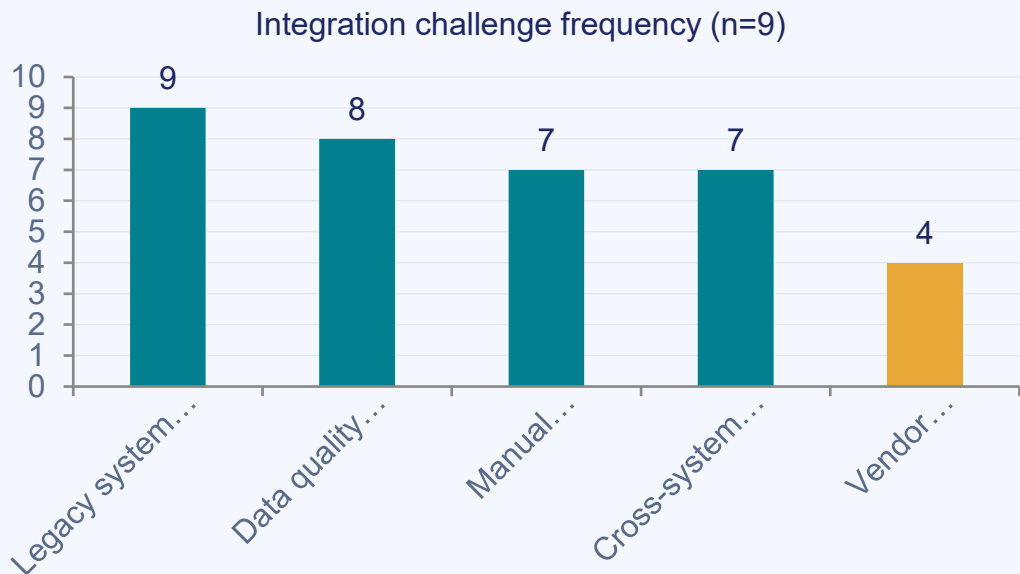
T5

Strategic Priorities & Path to Adoption

- Three-condition framework (original)
- User-centred design imperative
- Incremental, focused deployment

Theme 1 — Technical & Data Infrastructure Barriers

"From what we have observed, it relies heavily on spreadsheets. Even where EAM systems are in place, people are not comfortable using them." — CTO, Visualize Info



- ▶ One system per asset type structurally prevents cross-asset DT integration
 - ▶ Manual workarounds normalise inefficiency — harder to displace than the original technical problem
 - ▶ Bespoke connectors deployed to solve fragmentation entrench it further (vendor lockout)
- ⚠ Carmont case: unrecorded drainage asset → decade without inspection → fatal derailment 2020**

Theme 2 — The Abstraction Gap (original analytical contribution)

Abstraction gap: The differential comfort that different organisational actors have with abstract digital representations of physical assets — operating independently of digital literacy levels.

Senior managers



85% comfort with abstraction

High — abstract dashboards, KPIs, data tables

Asset engineers



55% comfort with abstraction

Medium — BIM models, 3D representations

Operational technicians



20% comfort with abstraction

Low — spatial, tactual, contextual thinking

Solution: LiDAR & photogrammetry interfaces — "a LiDAR scan is visually faithful to reality; everyone agrees it looks like what they are looking at" (CTO). User-centred design must be **architectural, not cosmetic**.

Theme 3 — Governance, Regulatory & Financial Constraints

Investment Paradox

£150,000 scanning equipment purchased centrally.

Operational resource to use it sits at regional level.

Regional resource competes with safety-critical maintenance budgets — which always take priority.

→ Capital acquired; value never realised.

Governance Accountability Vacuum

"The budget holder should not be in direct communication with the contractor, yet someone has to ensure the data gets uploaded." — AE

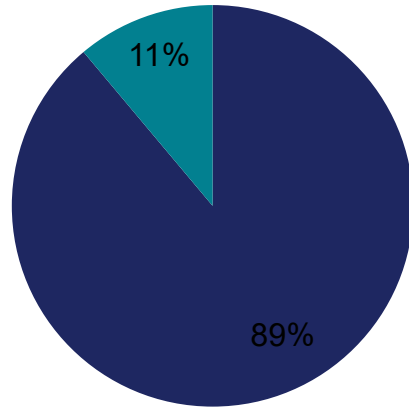
Result: Contractor responsible for uploading → contractor does not → data never entered → everyone moves on → Carmont.

Regulatory influence: constraint AND driver

Regulator	Role	Mechanism
ORR / RSSB	Constraint	Compliance burden; incident-driven workload increases
DfT	Driver	National Digital Twin Programme — principal policy enabler
CDM Regs	Dual	Exposes governance failures; drives reform through legal accountability (Carmont prosecution)

Themes 4 & 5 — Value Perception, Trust, and Strategic Priorities

Participant assessment of DT status (n=9)



■ Overhyped & underutilised ■ Appropriately valued

1. **Data first** 2. User-centred design 3. Risk-stratified deployment 4. Incremental demonstration

→ Strategic priorities feeding into NRDTAF

Trust finding

ALL 9 participants: qualified trust only
— DT as decision-support tool alongside engineering judgement, not standalone
— Trust located in the formal assurance process (product acceptance certification), not the technology itself

Three-Condition Framework (original construct)

Budget

Resource

Policy

Must be aligned simultaneously — not sequentially. Without all three, adoption drifts.

05 The NRDTAF — Network Rail Digital Twin Adoption Framework

P1 — Sequential prerequisite

Data First

Systematic data quality programme, governance reform, enforceable supply-chain accountability



P2 — Sequential prerequisite

User-Centred by Design

Interface fidelity to physical reality; design for operational users from the outset, not post-hoc



P3 — Sequential prerequisite

Risk-Stratified Deployment

Concentrate investment on high-frequency, high-impact assets where DT value is most demonstrable

▼ ADOPTION ZONE — Qualified Trust Threshold

P4 — Continuous condition

Three-Condition Alignment

Budget + resource + policy aligned throughout — time-bound policy targets prevent strategic drift

P5 — Continuous condition

Incremental Demonstration

Focused, problem-specific deployments generate visible evidence; build trust before scaling

06 Contributions to Knowledge & Conclusions



Empirical

Granular insider evidence from a safety-critical UK rail organisation — role-level, asset-class-level, and incident-level analysis including Carmont



Conceptual

The abstraction gap — a new theoretical construct not captured by TAM, DoI, or socio-technical theory. Operates independently of digital literacy



Analytical

The five-principle NRDTAF — operationally grounded framework integrating governance, design, risk, policy, and implementation from practitioner evidence

Four Principal Conclusions

- 1 DT adoption fails primarily at the governance level, not the technology level — regulation existed but was not enforced
- 2 The abstraction gap is the most underappreciated and practically consequential barrier in the literature
- 3 Incremental, evidence-based adoption is the realistic path — lift/escalator monitoring, ORBIS, BRAVE as proof of concept
- 4 The technology is not the problem — implementation immaturity is

Recommendations — Three Audiences

Network Rail

1. Data Governance Reform Programme as explicit precondition for DT deployment
2. Risk-stratified deployment strategy — high-frequency, high-impact assets first
3. User-centred design as non-negotiable procurement requirement
4. Time-bound digital adoption policy with named executive accountability
5. Formalise project-to-maintenance handover as a mandatory DT integration point

Policymakers & Regulators

1. Strengthen CDM enforcement for digital asset records (HSE + ORR review)
2. Commission national data standards for rail asset DTs via DfT / National DT Programme

DT Providers

1. Prioritise interface fidelity over feature comprehensiveness
2. Design for operational users, not just senior procurement decision-makers

Limitations & Directions for Future Research

⚠ Limitation 1 — Scope

Single-case design; findings transfer only to large, legacy-intensive, safety-regulated, publicly accountable organisations (Yin, 2018)

⚠ Limitation 2 — Sampling

No front-line maintenance technicians — the group most directly implicated by the abstraction gap. Their absence is the key evidence gap

⚠ Limitation 3 — Temporality

Cross-sectional; DT conditions evolve rapidly. Barrier conclusions retain validity; specific empirical details should be re-verified

F1

Longitudinal study

Track governance, data quality, and deployment outcomes as adoption progresses within Network Rail

F2

Comparative multi-organisation

Test abstraction gap concept and NRDTAF transferability across international rail operators and sectors

F3

Ethnographic front-line investigation

Capture how maintenance technicians actually interact with digital systems in the field (addressing sampling gap)

F4

Contractor data governance mechanisms

What enforcement structures reliably ensure as-built asset data is accurately recorded at project completion?

Digital Twin adoption at Network Rail is not a technology problem waiting for a technology solution.

It is a governance, accountability, and design problem — where the conditions for success must be deliberately constructed, not assumed to follow from investment alone.

Three original contributions: Empirical evidence · Abstraction gap concept · NRDTAF framework

Thank you — I welcome your questions.